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MILITARY SPECIFICATION

BATTERY CELLS AND ELEMENTS, LEAD-ACID, MAIN STORAGE, SUBMARINE; GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lead-acid main storage battery cells and elements for submarines.

1.2 Classification. Battery cells and elements shall be of the types specified below. Type shall be determined by the electrical capacity and configuration of the cell as specified by the applicable specification sheet (see 3.2 and 6.2.1). Types shall be as follows:

Type GUPPY I
Type GUPPY I, MOD C
Type GUPPY I, MOD D
Type TRIDENT

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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SPECIFICATIONS

FEDERAL

O-S-801 - Sulfuric Acid, Electrolyte; for Storage Batteries.
FF-W-92 - Washer, Flat (Plain).

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord Electrical, For Shipboard Use General Specification For.
MIL-C-915/6 - Cable, Electrical, 600 Volts, Types SHOF, DHOF, THOF, and FHOF.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, and Nuts.
DOD-C-2687 - Compound, Battery-Sealing (Metric).
MIL-T-10727 - Tin Plating; Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-C-15816 - Covers, Complete, and Parts, Hard Rubber for Use with Submarine Storage Batteries.
DOD-P-15817 - Parts, Synthetic Rubber, for Use with Batteries or in Battery Compartments of Submarines (Metric).
MIL-J-16128 - Jars, Battery, Laminated Rubber, Submarine.
MIL-A-22681 - Arrester, Flash, Main Storage Battery, (Submarine Use), General Specification For.
DOD-B-24541/1- Battery Cells and Elements, Lead-Acid, Main Storage, Submarine, Type GUPPY I (Metric).
DOD-B-24541/2- Battery Cells and Elements, Lead-Acid, Main Storage, Submarine, Type GUPPY I, MOD C (Metric).
DOD-B-24541/3- Battery Cells and Elements, Lead-Acid, Main Storage, Submarine, Type TRIDENT (Metric).
DOD-B-24541/4- Battery Cells and Elements, Lead-Acid, Main Storage, Submarine, Type GUPPY I, MOD D (Metric).
MIL-N-25027 - Nut, Self-Locking, 250°F, 450°F, and 800°F, 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/2- Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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2.1.2 Drawings and publications. The following drawings and publication form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5184219 - Submarine Storage Battery Curves and Data.

S6204-H-3, 211, 595 - Submarine Main Storage Battery Cell Jumpers.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S9086-G1-STM-010/CH-223-V1 - Naval Ships Technical Manual, Chapter 223, Lead-Acid Submarine Storage Batteries.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 276 - Stainless and Heat-Resisting Steel Bars and Shapes, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Battery cells and elements prepared under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.6).

3.2 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

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3.3 Electrical requirements.

3.3.1 Life. Life, in equivalent cycles or age, shall be as specified in the applicable specification sheet and shall meet the requirements of the guarantee (see 6.5).

3.3.2 Capacity. The capacity of each 126-cell or 128-cell battery based on an initial electrolyte temperature of 26.7 degrees Celsius (°C) shall be as specified in the applicable specification sheet. The average and final voltages specified in the applicable specification sheet reflect the voltage drop in the intercell connectors based on a two percent loss in the average voltage at the 1-hour discharge rate. The capacity for each of the various discharge rates of the applicable specification sheet is the rated capacity. The battery shall have a minimum of 90 percent of the 3-hour rated capacity within five cycles after electrolyte filling to demonstrate the initial capacity (see 4.7.2.4 and 6.4). The rated capacity at all the discharge rates specified in the applicable specification sheet shall be attainable within the number of equivalent cycles specified in the applicable specification sheet.

3.3.3 Cell terminals. The current density of cell terminal bus bar contact areas shall not exceed 38 amperes (A) per square centimeter (cm^2) at the 1-hour discharge rate.

3.3.3.1 Terminal resistance. The resistance between the terminal copper insert and the lead bus bar shall be less than 13 microhms when tested in accordance with 4.7.1.3.

3.3.4 Hydrogen emission. The hydrogen emission rate, at the time the cell reaches 90 percent of its rated capacity and during the service life routine (see 4.7.4.6) shall not exceed the value specified in the applicable specification sheet.

3.4 Physical requirements.

3.4.1 Dimensions. Cell dimensions shall be as specified in the applicable specification sheet.

3.4.2 Mass. In the "filled and charged" condition, the average cell mass and the minimum and maximum cell mass shall be within the values shown in the applicable specification sheet.

3.4.3 Cell jars. Unless otherwise specified (see 6.2.1), cell jars shall be in accordance with MIL-J-16128. The jars shall have a dielectric strength of not less than 100 megohms when tested in accordance with 4.7.2.2.

3.4.4 Cell covers. Unless otherwise specified (see 6.2.1), cell covers shall be designed and constructed in accordance with MIL-C-15816. The covers shall be designed to accept a flash arrester in accordance with MIL-A-22681 so installed in the vent opening that no leakage of gasses or electrolyte can occur around the edges. The flash arrester shall be the only cover fitting which will permit the passage of gasses from the cell. Service opening(s) equipped with gas-tight removable plug(s) shall be provided for watering and taking readings. Threaded apertures shall be provided on each side of the cover so that connection of the electrolyte circulating tube can be made. Any unused opening shall be sealed with a plug and soft rubber washer.

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3.4.5 Cover-to-jar-sealing. The cover-to-jar seal shall be battery sealing compound in accordance with DOD-C-2687.

3.4.6 Electrode pack assembly. Positive and negative electrodes shall be formed prior to final assembly into elements.

3.4.6.1 Grids. Positive grids shall be lead-calcium, lead-calcium-tin or lead-antimony alloy. Negative grids shall be lead-calcium or lead-antimony alloy. Positive and negative grids shall not exceed 0.085 percent calcium by mass. The grid material shall not contain impurities in quantity higher than as specified in table I. Grids shall be free of open sections, evidence of cold pouring, and blowholes. When lead burns are used, the resulting bond shall be uniform and free of obvious voids, cracks, and any foreign matter.

3.4.6.2 Active material. Active material of plates shall not have more breaks or loose parts than allowed by the manufacturer's inspection system plan (see 4.1.1). Active material shall not contain impurities in quantity higher than as specified in table I.

TABLE I. Maximum impurity concentration by mass in active materials and grids in parts per million (p/m).

Impurity	Concentration		
	Positive	Negative	Grid
Aluminum	50	50	5
Antimony	50	50	30 (See note 2)
Arsenic	1	1	20
Barium	1	No requirement	80
Bismuth	500	500	200
Cadimum	No requirement	No requirement	(See note 2)
Copper	50	50	50
Iron	100	100	50
Magnesium	50	50	50
Manganese	10	0.5	10
Mercury	0.1	0.1	0.1
Nickel	1	1	0.1
Platinum	0.1	0.1	0.1
Silver	10	10	10
Strontium	10	10	50
Tin	10	10	(See note 3)
Zinc	10	10	5

NOTES:
1. The sum of all other impurities shall not exceed 50 p/m with no individual impurity exceeding 5 p/m.
2. Antimony and cadimum are not contaminants for cell type GUPPY I, MOD D (see 1.2.1).
3. Tin is not a contaminant, but rather, a grid alloying agent, and can be found in varying amounts depending upon the various contractor's casting methods and manufacturing processes.

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3.4.6.3 Plate tab-to-bus bar bond. The bond between the plate tab and the bus bar shall offer an electrical resistance of less than 100 microhms when tested in accordance with 4.7.2.1.

3.4.6.4 Physical characteristics. The electrodes shall fall within the limits of tolerances established at the time of qualification approval for the applicable type. The physical characteristics of the electrode pack assembly shall be as described for the qualification test samples and shall be submitted prior to authorization of qualification tests (see 4.3.2 and 6.2.2). They shall include the following:

(a) Density of active material (after electrochemical formation and drying of the paste).
(b) Electrode dimensions, including thickness.
(c) Mass of active material per electrode and gross mass of electrodes.
(d) Number of positive and negative electrodes.
(e) Separator system description from positive electrode to negative electrode (see 3.4.7).
(f) Separator characteristics in terms of thickness, porosity, average pore size, resistivity, and analysis (see 3.4.7).

3.4.7 Separators. Separators of microporous rubber shall be used. Fiberglass mats and perforated sheet retainers may be used in combination with microporous rubber separators. Vertical binding strips are required on both positive and negative electrodes. The characteristics of the microporous rubber separator shall be as follows:

Porosity: 50 to 65 percent.
Electrolyte resistivity in sulfuric acid of 1.280 specific gravity: 2.5 to 6.4 ohm/cm.
Moisture content by mass: not to exceed 2 percent in a maximum of 75 percent relative humidity at a maximum temperature of 35°C.
Impurities: not to exceed the following limits (p/m).

Aluminum	1000
Chloride (as Cl)	70
Copper	30
Iron	500
Magnesium	500
Manganese	10

The sum of all other metallics shall not exceed 80 p/m.

3.4.8 Terminals. Terminal posts and bus bars shall be lead-coated copper. Terminal intercell connector bolts, nuts, and washers shall be stainless steel in accordance with ASTM A 276, type 316. Bolts shall be in accordance with MIL-S-1222, nuts in accordance with MIL-N-25027 and figure 1, and washers in accordance with FF-W-92. Lead coating on parts not exposed to direct action of electrolyte shall be applied by electroplating and shall be at least 0.05 millimeters (mm) thick. Copper parts that are exposed to the direct action of electrolyte shall be protected by a coating of lead or lead alloy of at least 6 mm thickness. Terminal posts shall be marked as specified in 3.4.14.

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3.4.8.1 Cooling water channels. The cells and elements shall be provided with channels through the terminal posts and straps for the circulation of cooling water when required by the applicable specification sheet. Plugs shall be supplied for each channel opening. Each terminal post and strap assembly shall show no leaks when tested in accordance with 4.7.1.2. The construction of the channel in the terminal posts and plate straps shall be such that cooling water in passing through the channel shall not be in contact with lead or lead alloys. The inside of water cooling channels shall be tin plated with a minimum thickness of eight micrometers in accordance with MIL-T-10727, type II or equal. The preservative and salt spray tests shall not be required.

3.4.9 Intercell connectors and jumpers. The intercell connectors, which are provided by the battery installing activity, shall be lead-coated copper with a coating of 0.08 mm thickness. The connector (or jumper), bolts, and nuts shall be tightened to a torque of 57-62 newton-meters. The cell assembly shall withstand this torque without damage. Cell jumpers shall be supplied when required by the contract (see 6.2.1). The length of cell jumper assemblies shall be approximately 0.75 meter (m) and 1.50 m for GUPPY's, and 1.80 m for TRIDENT in accordance with Drawing S6204-H-3, 211, 595, of extra flexible rubber covered cable, type SHOF conforming to MIL-C-915 and MIL-C-915/6. Between any two jumpered terminal posts, the jumper assembly shall be not less than 760 square millimeters in area. Terminal adapters shall be used to interconnect individual cell terminal posts.

3.4.10 Rubber parts. Soft rubber parts shall be in accordance with DOD-P-15817. Hard rubber parts (except jars, separators, sheathings, and baffles) shall conform to MIL-C-15816.

3.4.11 Electrolyte.

3.4.11.1 Electrolyte composition. Sulfuric acid electrolyte shall be in accordance with O-S-801. Electrolyte specific gravity, when fully charged, shall be as specified in the applicable specification sheet.

3.4.11.2 Electrolyte impurities. The amount of impurities in the electrolyte of a fully charged cell shall not exceed the limits specified in table II at the time the battery reaches its initial capacity.

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TABLE II. Maximum impurity concentration in electrolyte.

Impurity	Determined as	Limit in (p/m)
Ammonium	NH_4	220
Antimony	Sb	10
Arsenic	As	1
Chlorine (including perchlorates)	Cl	20
Copper	Cu	20
Iron	Fe	60
Manganese	Mn	0.6
Nickel	Ni	0.5
Nitrates	NO_3	20
Platinum	Pt	0.1
Suspended matter	—	Lead compounds only

3.4.11.3 Electrolyte agitator. The cells and elements shall be provided with an electrolyte agitating tube or tubes designed so that an air supply from outside the cell can be brought into the tube or tubes approximately 380 mm below the collar of the air supply tube. The minimum pumping rate of 15 liters (L) of electrolyte per hour shall be obtained when air is applied at a rate of 850 milliliters per minute per tube and electrolyte is at minimum height. The connection to the electrolyte pumping tube or tubes shall extend through either of two apertures in the cover. The contractor shall furnish gas-tight and acid-tight connections to the cell or element.

3.4.12 Flash arresters. The flash arresters shall be in accordance with MIL-A-22681.

3.4.13 Assembly of cells and elements. Cells and elements (see 6.7.2 and 6.7.3) shall be free of lead droppings or run downs between the plates that could cause cell failure. The cells shall be constructed to permit the installation or removal of an element without the use of lead burning. The elements shall be inserted in the jars and the covers sealed in accordance with the procedures specified in NAVSEA S9086-G1-STM-010/CH-223-V1. Assembled cells shall withstand an internal air pressure of 7 kilopascals (kPa) for 1 minute without leakage or drop in pressure when tested in accordance with 4.7.1.1.

3.4.14 Marking of parts. Elements shall be marked with a serial number, contract year, and the battery manufacturer's name. The battery manufacturer's name shall not be marked on the cell cover. Hard rubber parts shall be molded with a piece number or letter for identification. Positive terminal posts shall be marked "POS". Negative terminal posts shall be marked "NEG". The terminal connecting strap shall be marked in accordance with 4.7.1.3.

3.4.15 Mercury. Mercury shall not be used in the manufacture or in any test or test equipment in which there is any danger of introducing mercury into the battery through accident or otherwise.

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3.4.16 Raw materials. Unless otherwise specified herein, only primary corroding or virgin lead shall be used for the production of positive oxide or positive grids.

3.4.17 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.5 Environmental requirements.

3.5.1 Temperature. Cells or elements shall not be adversely affected by environmental temperature of minus 18 to 55°C in storage (dry condition) or 0 to 55°C after installation (wet condition).

3.5.2 Shock. Cells shall conform to the requirements for grade A, class I, type A, medium weight equipment specified in MIL-S-901.

3.6 Deliverable hardware.

3.6.1 Sets of cells or elements. The number of sets as defined in 6.7.1 shall be as specified (see 6.2.1). For cells with one service tube, each set shall contain 50 percent cells of the right hand configuration and 50 percent cells of the left hand configuration. Cells or elements shall be delivered formed and dry.

3.6.2 Terminal parts. Cells and elements shall be supplied complete with terminal intercell connector nuts, bolts, and washers. Elements shall be supplied with 20 percent of seal nuts.

3.7 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2). Technical manual content requirements shall include, but not be limited to, the following:

(a) Battery curves and data drawing shall be in the format of Drawing 803-5184219.
(b) Manual content shall be in accordance with NAVSEA S9086-G1-STM-010/CH-223-V1, making reference to this technical manual, wherever applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Verification of qualification (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the examination specified in 4.6 and the tests specified in 4.7 through 4.7.4.6

4.3.1 Sample cells for qualification inspection. Seven sample cells of the type for which qualification is desired shall be submitted in the dry condition with instructions for filling and establishing initial capacity (see 4.7.2.4).

4.3.2 Authorization for qualification tests. Prior to authorization of qualification tests, the data specified in 3.4.6.4 and 3.7 shall be submitted to and accepted by NAVSEA.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sample cells or elements shall be selected for quality conformance inspection as required by the contract or order (see 6.2.1). Three cells or elements installed in standard jars are required. The cells or elements shall be selected from the first set of cells or elements and from every third set of the same type thereafter (4th set, 7th set, 10th set, and so forth). The number selected shall be three cells or elements per lot of three sets of cells or elements to be delivered. When the sampling interval results in selection of a sample from the last set under current production on a contract or order, the contractor may elect that the sample be taken from the next to the last set instead.

4.4.2 Inspection. Sample cells selected in accordance with 4.4.1 shall be subjected to the examination and tests specified in 4.6, and 4.7.1 through 4.7.2.7. Any test cells or elements, as well as the set represented by the sample, which fail to pass the required tests, shall be rejected. Shipment of the set may be authorized by the Defense Contract Administration Service Management Area (DCASMA) upon successful completion of tests specified in 4.7.2.3 through 4.7.2.7, pending results of subsequent tests.

4.5 Verification of qualification. The Government reserves the right to select any cells from any delivered lot for verification of qualification which shall be conducted in a Government laboratory, and shall consist of the examination specified in 4.6 and the tests specified in 4.7 through 4.8.

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4.6 Examination. Parts used in the assembly of the cells shall be examined for conformance to the requirements of this specification and to the contractor's quality control provisions. Assembled cells and elements shall be examined for any visible physical defects and for conformance to the requirements of this specification which do not involve tests. Any defective cell or element shall be rejected.

4.7 Tests.

4.7.1 Tests for all cells or elements manufactured.

4.7.1.1 Pressure-test, cover-to-jar seal. All assembled cells shall be air-pressure tested at 7 kPa for 1 minute through an opening of the cover with all other openings sealed and shall meet the requirements specified in 3.4.13.

4.7.1.2 Terminal posts and strap assembly. For terminals employing water cooling channels sampling for terminal posts and strap assemblies shall be in accordance with MIL-STD-105 and shall have an AQL of 1 percent. The sample(s) terminal post and strap assembly shall be tested under water with not less than 210 kPa air pressure to determine conformance with 3.4.8.1.

4.7.1.3 Terminal resistance. The contact resistance between the terminal copper insert and the bus bar shall be measured in accordance with figure 2. Each terminal copper insert shall be measured separately. All terminal assembly connecting straps which meet the requirements of 3.3.3.1 shall be stamped with the letter "O". The values obtained shall not exceed the requirement specified in 3.3.3.1.

4.7.2 Tests on sample cells or elements.

4.7.2.1 Plate-to-bus bar bond. The bond between each plate tab and the bus bar shall be tested for electrical resistance with a microhmeter on all sample cells or elements. Measurements taken from on top of the bus bar to on top of the electrode on either side of the bus bar shall meet the requirements specified in 3.4.6.3.

4.7.2.2 Cell for dielectric strength. When required by the contract or order (see 6.2.1), each sample cell shall be submitted to the dielectric strength measurement in accordance with figure 3. The dielectric strength of the sample cells shall meet the requirements of 3.4.3.

4.7.2.3 Mass of complete cells. The sample cells shall be weighed individually with the electrolyte at normal level at the end of a charge. Cell connectors and terminal bolts shall not be included. The mass shall be within the specified maximum and minimum cell mass requirements in the applicable specification sheet.

4.7.2.4 Electrical tests. After filling of the sample cells, a maximum of five cycles of charge and discharge in accordance with the manufacturer's preferred procedure is allowed to develop capacity prior to the tests of 4.7.2.4.1 and 4.7.2.4.2. All subsequent tests shall be run in an ambient of 25 + 10°C with capacity measurements corrected to an electrolyte temperature of 26.7°C.

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4.7.2.4.1 Low rate initial capacity. The cells shall be discharged at the 3-hour rate to the cut-off voltage and shall give at least 90 percent of the rated capacity after correction for temperature.

4.7.2.4.2 High rate initial capacity. After recharge, the cells shall be discharged at the 1-hour rate to the cut-off voltage and shall give at least 80 percent of the rated capacity after correction for temperature.

4.7.2.4.3 Failure on electrical tests. Failed cells may be retested within the number of cycles of any charge or discharge of 4.7.2.4 allowed after filling with electrolyte, if a maximum number of seven cycles has not been reached.

4.7.2.5 Hydrogen emission. While at the finishing rate of an equalizing charge conducted in accordance with the battery technical manual, the rate of hydrogen emission from each cell shall be measured at 15-minute intervals and recorded. After completing the equalizing charge, the cells shall be placed on open circuit and the rate of hydrogen emission from each cell shall be measured at 15-minute intervals during the first hour and hourly for 12 hours thereafter and recorded. Initial open circuit readings shall commence within 1 minute after start of the 13-hour open circuit period. The average rate of hydrogen emission expressed in cubic centimeters (cm^3)/hour/cell/1000 ampere hours (Ah) at the 10-hour rate and reduced to 0°C and 1.0 atmosphere pressure shall meet the requirements specified in the applicable specification sheet.

4.7.2.6 Electrolyte analysis. At the end of the tests, after a charge, a sample of 1 L of electrolyte shall be drawn from each cell. The samples shall be analyzed for impurities. The amounts of impurities shall not exceed the limits specified in 3.4.11.2.

4.7.2.7 Demonstration of rated capacity. After the sample cells have met the tests specified in 4.7.2.4 through 4.7.2.6 they shall be cycled to demonstrate their rated capacity within the number of cycles specified in the applicable specification sheet. The rated capacity shall be demonstrated at the 1-hour, 3-hour, and the 10-hour discharge rate. If the three cells equal or exceed the rated capacity within the specified cycles, they will be considered as representing all the cells of the same type furnished in the contract or order. If they fail to give the required capacity, the group of three sets of cells or elements from which they are taken shall be subject to rejection. At the option of the contractor, sets of cells or elements may be shipped prior to completion of the capacity test. However, if upon completion of this test, the capacity has not been met, cells or elements already shipped shall be subject to rejection.

4.7.3 Shock test. One charged cell, filled with electrolyte shall be tested to determine conformance with 3.5.2. The cell shall be mounted in a manner duplicating shipboard installation as nearly as possible, but at an angle such that the plane of the cell plates is parallel to a plane tilted down 45 degrees from the vertical. The cell shall be delivering current at a constant rate between the 3-hour and 10-hour discharge rates during testing. Voltage and current shall be continuously recorded for a time interval lasting from not less than 15 seconds after the last shock blow until the cell has discharged to the specified minimum final cell voltage. Ninety percent of the rated capacity at the discharge rate used shall be demonstrated.

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4.7.4 Life test. Life test regimes shall be as required by the applicable specification sheet. Each cell shall be maintained during the entire life test as recommended in the service manual. The electrolyte agitators shall be continuously operated by supplying 850 cm^3 of air per minute per tube. When a cooling system is required, cooling water shall be circulated at a temperature of 35 $\pm$ 5°C at a rate of 1,500 cm^3 per minute per cell through the channels provided in the terminal posts and straps. Three sample cells shall be connected in series and operated as a three cell battery for each of the separate life test regimes required by the applicable specification sheet. All tests shall be run at an ambient temperature of 25 $\pm$ 10°C.

4.7.4.1 Accelerated float charge regime. One 3-cell battery shall be subjected to a repetitive 21-day routine as follows:

(a) On each of the five days in a work week, discharge the battery at the 10-hour rate for 5 hours.
(b) On each of the first four days of the work week, recharge the battery for 19 hours at the float charge voltage recommended in the battery technical manual.
(c) On the last day of the work week, the battery will be recharged for 66 hours at the float charge voltage recommended in the battery technical manual.
(d) Repeat (a) through (c) until a total of 13 cycles have been completed.
(e) Upon completion of the recharge of each 13th cycle of each routine, the battery shall be given a full capacity test discharge at the 3-hour rate to the minimum cell voltage (average or low individual cell) as shown in the applicable specification sheet.
(f) The battery shall be recharged for a minimum of 69 hours at the float charge voltage recommended in the battery technical manual.
(g) The 14-cycle routines shall be repeated, continuously, until the capacity falls below 80 percent of the rated value at the 3-hour rate, corrected for temperature.

4.7.4.2 Ship float simulation regime. One 3-cell battery shall be subjected to a repetitive 13-week routine as follows:

(a) On the first day of each 7-day period, the battery shall be discharged at the 20-hour rate for 1 hour and recharged, for 23 hours, at the float charge voltage recommended in the service manual.
(b) On the second and third day of each 7-day period, the battery shall be discharged at the 10-hour rate for 1 hour and recharged, for 23 hours, at the float charge voltage recommended in the service manual.
(c) On the fourth day of each 7-day period, the battery shall be discharged at the 48-hour rate for 1 hour, discharged at 1/5 of the 48-hour rate for 5 additional hours and recharged, for remainder of the 7-day period at the float charge voltage recommended in the service manual.
(d) Repeat 4.7.4.2 (a) through 4.7.4.2 (c) for 11 7-day periods.
(e) On the first day of the 13th 7-day period, the battery shall be given a full capacity test discharge at the 3-hour rate to the minimum cell voltage specified in the applicable specification

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sheet and recharged, for the remainder of the 7-day period, at the float charge voltage recommended in the service manual.

(f) The test routine from 4.7.4.2 (a) through 4.7.4.2 (e) shall be continuously repeated until the capacity drops below 80 percent of the rated capacity at the 3-hour rate.

4.7.4.3 Cycling regime. Two 3-cell batteries shall be subjected to a repetitive 56-day routine as follows:

(a) For the first four days of each 7-day period, discharge the battery at 195 A (GUPPY size cell) for 17 hours.
(b) Each discharge shall be followed by a normal charge as specified in the battery technical manual.
(c) Each cycle (discharge and charge) shall not exceed 24 hours.
(d) The battery shall be placed on open-circuit for the remainder of each 7-day period.
(e) A routine shall consist of 32 cycles and will require 56 days to complete.
(f) The 16th cycle of each routine shall contain an (N-1) charge. This is a normal charge which has been extended for one hour at the finishing rate specified in the battery technical manual.
(g) The 32nd cycle of each routine will contain an equalizing charge as specified in the battery technical manual.
(h) Upon completion of each odd-numbered routine, the battery will be given a capacity-measuring discharge of the 10-hour rate, followed by an equalizing charge as shown in the battery technical manual.
(i) Upon completion of each even-numbered routine, the battery will be given a series of four capacity-measuring cycles.
(j) The discharges will be performed in the following order: 10-hour rate, 1-hour rate, 48-hour rate and the 1/2-hour rate.
(k) Each discharge in (j) above shall be followed by an equalizing charge as specified in the battery technical manual.
(l) The service life test procedure from (a) through (k) shall be continuously repeated until the A hour capacity, at any approved hourly rate, falls below 80 percent of the rated capacity.

4.7.4.4 Battery failure procedure. When battery capacity drops below 80 percent of the rated capacity at the 3-hour test discharge rate during the tests of 4.7.4.1 and 4.7.4.2, battery capacity shall be restored by the procedure defined in the service manual. Testing shall continue after successfully completing the capacity restoration procedure.

4.7.4.5 Life test failure. The cell type (see 1.2.1) shall be considered to have failed the life test when, during either of the tests of 4.7.4.1 or 4.7.4.2, the failure procedure of 4.7.4.4 is required after consecutive capacity test discharges, or is required more than three times before the requirements of the guaranteed service life, in terms of equivalent cycles, as specified by the applicable specification sheet is met.

4.7.4.6 Hydrogen emission. For the regimes specified in 4.7.4.1, 4.7.4.2, and 4.7.4.3, the hydrogen emission shall be measured in accordance with 4.7.2.5 semi-annually. The emission rate shall not exceed the value specified in the applicable specification sheet.

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4.8 Criteria of acceptance for qualification. The cell type (see 1.2) shall be accepted as qualified upon conformance to the qualification inspection requirements of 4.3, and upon meeting the requirements of the guaranteed service life in terms of equivalent cycles as specified in the applicable specification sheet at the completion of life tests (see 4.7.4).

4.9 Inspection of packaging. The cleanliners, preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.8.)

5.1 Cleanliness, preservation - packaging, packing, and marking. Cleanliness, preservation-packaging, packing, and marking shall be level A as specified herein.

5.2 Cleanliness (level A). Cleanliness of materials used for packaging and packing shall be sufficient to prevent the introduction of impurities (see table I) into cells or elements.

5.3 Preservation-packaging, and packing (level A).

5.3.1 Cells and elements. Cells and elements shall be assembled for shipment in packing containers required by NAVSEA. Containers shall be of the overseas type or class and designed to handle each cell in either the "filled and charged" or "uncharged and dry" condition. Elements shall be supported and secured in a manner that will prevent damage to elements and components thereof. Plates, separators, and components shall be secured in a manner that will prevent plate spread movement of separators or other components. Cells and elements shall be packaged and loaded with the plates perpendicular to the direction of motion.

5.3.1.1 The following information shall be sealed in a waterproof bag or envelope and packed in the first packing container and each additional 50th packing container:

(a) A copy of the approved Curves and Data drawing shall be furnished for cells in the "filled and charged" condition.
(b) A copy of the approved Curves and Data drawing and approved instructions for putting into service shall be furnished for cells or elements in the "dry and uncharged" condition.

5.4 Marking (level A). Packing cases for cells and elements shall be marked in accordance with MIL-STD-129. In addition, each packing case shall be clearly marked and numbered to facilitate ready identification regardless of stowage position. Markings on the exterior of packing cases shall include:

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(a) Submarine main storage batteries.
(b) Contractor's name.
(c) Contract number.
(d) Item and set number, as applicable.
(e) Box serial number with total number of boxes comprising the set or item as applicable.
(f) A brief identifying description of the material in the containers such as "Cells, Main Storage Battery" "Dry and uncharged" or "Wet and charged", "Spare cell" or "Spare element".
(g) Contractor's type designation.
(h) National stock number (furnished by NAVSEA upon request).
(i) Mass and cubage.
(j) "FRAGILE." "HANDLE WITH CARE." "THIS END UP".
(k) Mark crate number 1 as follows: "CONTAINS MANUALS" (see 5.3.2).

5.5 Technical data (see 3.7). When specified (see 6.2.1 and 6.2.2), packaging-packing drawings shall be submitted. These drawings are not required when previously submitted and accepted.

6. NOTES

6.1 Intended use. The lead-acid main storage battery cells and elements are intended for use in submarines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification and applicable specification sheet.
(b) Type required (see 1.2).
(c) Cell jar and cover material if other than as specified (see 3.4.3 and 3.4.4).
(d) When cell jumpers shall be required (see 3.4.9).
(e) Number of sets of cells or elements (see 3.6.1).
(f) Which sample cells or elements shall be selected for quality conformance inspection (see 4.4.1).
(g) When dielectric testing of sample cells is required (see 4.7.2.2).
(h) When packaging technical data is required (see 5.5).
(i) If engineering services are required (see 6.3).
(j) Guarantee as specified (see 6.5).
(k) When installation and checkout repair parts and tools are required (see 6.9.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.4.6.4	Test report	DI-T-2072	10.1.b
3.7 and 5.5	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - contractor Drawing no. - contractor
3.7	Manual, technical standard	DI-M-2044	MIL-M-15071, Type I
4.1.1	Inspection system program plan	DI-R-4803	----
6.3	Report, contractor engineering and technical services (CETS)	UDI-E-23127	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Engineering services (see 6.2.1). The services of the contractor's engineer should be available upon the Naval Sea Systems Command's (NAVSEA) request in order to:

(a) Assist the shipyard in preparing the battery for installation and installing it.
(b) Assist the ship's force in becoming familiar with the battery.
(c) Examine the battery condition and take corrective action in case of unexpected problems.
(d) Participate in technical meetings with NAVSEA or the ship's force when involving the batteries and in locations other than the manufacturer's plant.

An average of 15 mandays should be provided for the entire life of each battery installed. The maximum number of mandays should be as specified in the contract or order for the entire number of batteries which can be assembled from cells or elements bought on the same contract. A field engineering report should be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

6.4 Bids. The provisions of 4.7.2.7 for frequency of performance of the service rating capacity test are such that most large contracts will require

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performance of this test at some time during the contract. In order to provide a consistent basis for comparison of bids, bids should be based on the assumption that one service rating capacity test per type will be required. If deliveries are required to extend over a period longer than 12 months, it should be assumed that two service rating capacity tests will be required.

6.5 Guarantee. The guarantee shall be as specified in 6.5.1 through 6.5.4 (see 6.2.1).

6.5.1 Nothing set forth in this guarantee provision shall limit any rights afforded to the Government by the inspection clause relating to latent defects, fraud, and gross mistakes that amount to fraud. Not withstanding inspection and acceptance by the Government of supplies furnished under the contract or any provision of the contract concerning the conclusiveness thereof, the contractor guarantees the life of the cells and elements furnished on this contract in terms of time under service operating conditions. Cells or elements furnished shall be guaranteed for the months or equivalent cycles specified in the applicable specification sheet.

6.5.2 The period of guarantee for cells or elements should be considered to begin on the date the battery is put into service (start of initial charge), except when the time between the date of shipment and the date of initial charging exceeds two years, in which case the guarantee should begin two years from the date of shipment. The life of the cells or elements should be considered at an end when the cells cannot deliver 80 percent of their rated capacity, when handled and maintained in accordance with the methods and procedures specified by the contractor's service manual. Defective cells may be jumpered out of the battery so long as the remaining cells are capable of delivering 80 percent of the rated capacity (including any correction for temperature), before either the total battery voltage or any individual cell voltage falls below its cut-off voltage, at the corresponding discharge rate, in accordance with the applicable specification sheet. In case of failure to meet the guaranteed life, it is understood and agreed that the battery manufacturer will indemnify the Government on a prorata basis (proportionately) for equivalent cycles or months of service not obtained. (If P is the purchase price of the cells or elements, n the number of months not obtained, Q the number of equivalent cycles not obtained, m the months of guaranteed service life in the applicable specification sheet, and E the number of guaranteed equivalent cycles in the applicable specification sheet, the smaller of the two numbers $P \times n/m$ and $P \times Q/E$ should be used.)

6.5.3 At the option of the Government, the amount to be paid the Government may be assessed as a cash indemnity, or may be applied to reduce the cost of any other similar submarine main storage battery or set of elements which the Government may contract to purchase from the contractor. It is understood and agreed that a cash indemnity will only be assessed when, in the opinion of the Government, experience indicates that the type of cell manufactured under this contract by the contractor is unsuitable for use as submarine main storage batteries.

6.5.4 With respect to sets of cells or elements from which no sample was taken for factory acceptance tests prior to shipment, or with respect to sets which the Government may elect to accept even though samples were tested which failed, the manufacturer further guarantees that each such set furnished will,

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when filled and given an initial charge by the Government or its agent in accordance with manufacturer's instructions, yield 90 percent of the rated capacity at the 3-hour discharge rate within the first five cycles (exclusive of those allowed for the development of the initial charge). For this purpose, a cycle consists of a discharge at one or more rates from a fully charged condition to the low voltage limit for the rate of discharge being used, together with the following charge which brings the battery back to the condition which existed at the beginning of the discharge. In the event the specified initial capacity is not obtained within the said five cycles, the manufacturer agrees to pay the Government the amount equal to all reasonable costs incurred by the Government for cycling beyond five cycles as necessary to develop the specified initial capacity. The manufacturer further agrees that use of premium labor, services or materials for such additional cycling will not be the basis for dispute of the amount of said costs, if such is either the normal practice of the activity performing the work, or is required to expedite preparation of the battery so as to not interfere with other work.

6.6 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24541 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.6.1).

6.6.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.7 Definitions. The definitions specified in 6.7.1 through 6.7.3 are applicable to this specification.

6.7.1 Set. A set is the total number of cells or a total number of elements sufficient to assemble one complete battery installation in a ship.

6.7.2 Element. An element is a complete assembly consisting of the positive group and the negative group of electrode assemblies mounted together with separators and retainers in place, in all respects ready for installation in the jar.

6.7.3 Cell. A cell is an element installed in a jar with cover sealed in place and other hardware required by this specification installed. There are two types of cell configurations for cells with one service tube: right and left hand. The right hand configuration is shown in the plan view of the cell cover in the applicable specification sheet. The left hand configuration interchanges the position of the service tube and the cell marker.

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6.8 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.9.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified. When required by the contract or order, repair parts and tools for each set of 126 or 128 cells should be ordered in the amounts indicated by table III.

TABLE III. Installation and checkout repair parts and tools.

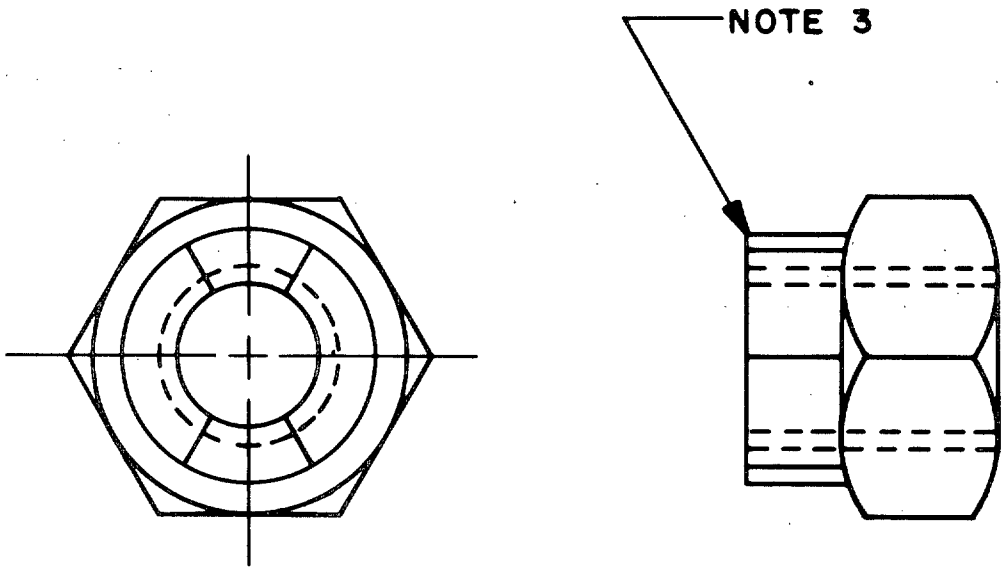
Quantity		Distribution		
Name of part		Number supplied	Installation and repair parts	Onboard repair parts
Flash arrester assembly		12	2	10
Hard rubber seal nuts		20	10	10
Hard rubber filling plugs		15	5	10
Complete sets of soft rubber gaskets		15	5	10
Electrolyte agitation air supply tube:				
GUPPY types		131	5	126
TRIDENT type		266	10	256
Special agitation air supply tubes for GUPPY types, long		10	--	10
Sets of electrolyte agitation fittings:				
GUPPY types		10	5	5
TRIDENT type		20	10	10
Sets of wrenches		2	1	1
Electrolyte height indicators		2	--	2
Cell jumper cable assembly:				
GUPPY I type:	0.75 m	16	--	16
	1.50 m	8	--	8
GUPPY I, MOD C and D types:	0.75 m	8	--	8
	1.50 m	4	--	4
TRIDENT type:	1.80 m	24	--	24

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6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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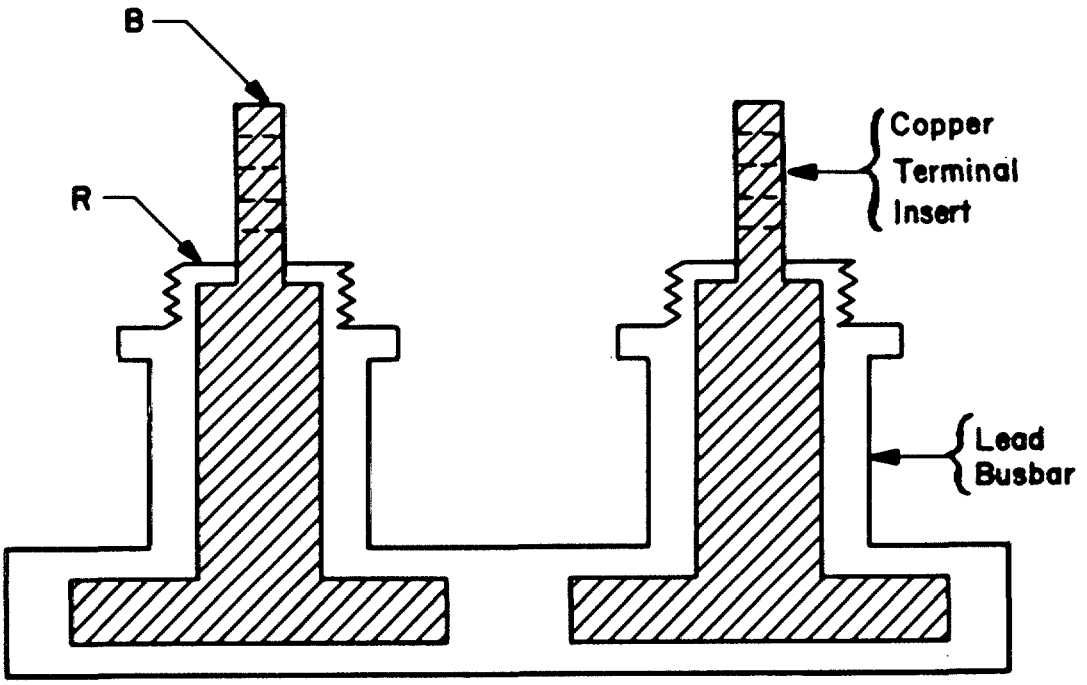
SH 11271

NOTES:

1. Nut threads shall be coated with molybdenum disulfide dry film lubricant.
2. Identification: "316" stamped on flat of hex; multiple stamping permissible.
3. Six slots equally spaced collar crimped to form locking feature to meet torque specifications of MIL-N-25027.

FIGURE 1. Self-locking nut, 250 degrees Fahrenheit (°F).

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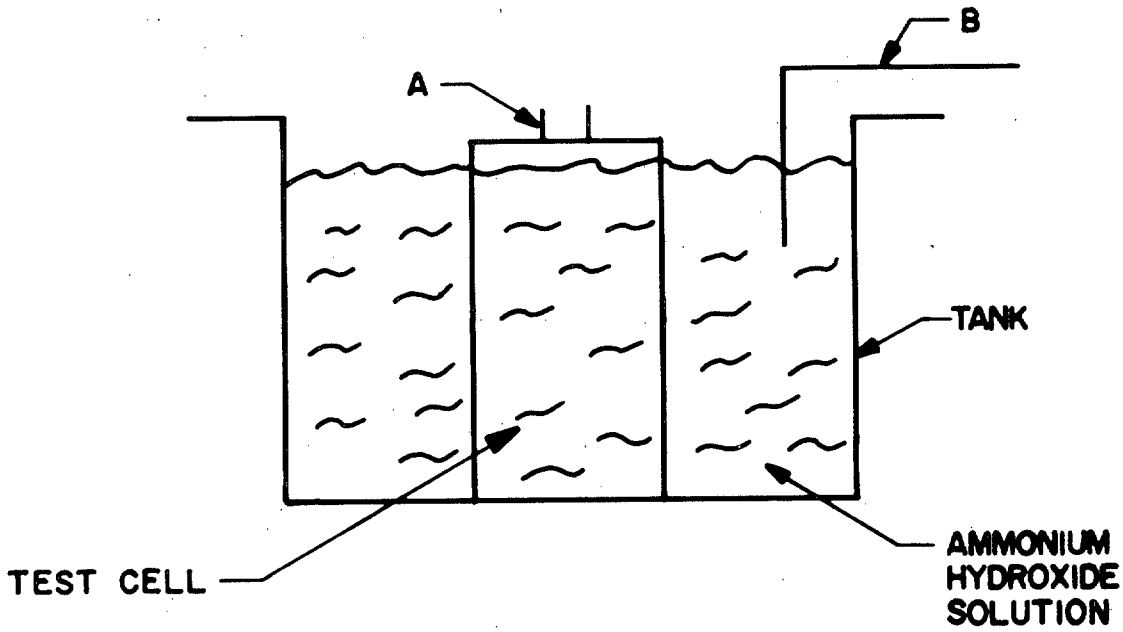


SH 11270

Measure electrical resistance at point R and B on each terminal of the positive and negative bus bar.

FIGURE 2. Terminal resistance measurement.

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SH 12117

Measure dielectric strength at
points A and B

FIGURE 3. Cell dielectric strength measurement.

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